

# E.KARTHIK

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## SUMMARY

**Full Stack & AI Engineer** with hands-on experience building production-ready web applications, autonomous AI agents, and hybrid GraphRAG pipelines. Proven ability to ship end-to-end systems spanning REST APIs, vector/graph databases, and LLM-powered automation.

## EDUCATION

- **B.Tech** in Computer Science & Engineering — Mohan Babu University CGPA: 8.95 2023 - 2027
- **BIEAP** — Physics, Chemistry & Mathematics — Sri Chaitanya College CGPA: 9.16 2021 - 2023
- **SSC** — General — Sri Venkateswara School CGPA: 10.0 2020 - 2021

## TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript
- **GenAI / AI Engineering:** RAG, GraphRAG, LLM Applications, Embeddings, LangChain, LangGraph, Gemini SDK, Prompt engineering, Ollama, Hugging Face, Unsloth, Transformers, Deep Learning, Agentic AI(crewai)
- **Frontend:** HTML, CSS, React, Bootstrap, Tailwind CSS
- **Backend:** Node.js, Express.js, REST APIs, EJS
- **Databases:** MongoDB, MySQL, Neo4j, Pinecone
- **Tools:** Git, GitHub, Streamlit, Hoppscotch, Docker, Google Colab, Claude, CrewAI

## WORK EXPERIENCE

Wanderlust – Travel Listing Web Application | [Live Demo](#)

- **Engineered** a production-ready travel listing platform using a modular MVC architecture, enabling full CRUD operations for managing travel destinations with **Node.js**, **Express.js**, **MongoDB**, **Mongoose**, and **EJS**.
- **Implemented** secure user authentication and authorization using **Passport.js**, along with server-side validation using **Joi**, ensuring protected access and maintaining data integrity.
- **Integrated** an **AI-powered chatbot** to enhance destination discovery and provide intelligent travel assistance, while organizing the application with structured routing and controllers for scalability and maintainability.

## PROJECTS

Movie Search Engine | [Live Demo](#)

- Engineered a **hybrid GraphRAG** pipeline using **LangChain**, integrating **Pinecone** vector embeddings with **Neo4j** knowledge graphs to enable context-aware retrieval across a dataset of **1,000+ movies**.
- Built a **REST API** Leveraging the **Google Gemini SDK** for embedding generation, semantic search, and LLM-powered question answering, with **pdf-parse** and vector indexing for unstructured data retrieval.
- Integrated **Model Context Protocol** to enable real-time tool-based retrieval of **crypto**, **live news** and **weather** data, combining live external information with structured and semantic knowledge retrieval.

Autonomous AI Website Builder | [code](#)

- Developed a **CLI-driven** autonomous Generative AI agent that converts natural language user requirements into responsive frontend websites using HTML, CSS, and JavaScript.
- Integrated the **Google GenAI SDK** to generate structured code through **LLM-powered** prompt engineering and iterative code generation.
- Automated the development workflow by leveraging **Node.js** modules such as **child\_process** for command execution, **os** for system operations, **util** for asynchronous utilities, and **readline-sync** for interactive CLI input.

Automated Code Refactoring Agent | [code](#)

- Built an autonomous AI-powered code auditing agent in **Node.js** that securely scans code directories, patches software vulnerabilities, and refactors logic bugs.
- Integrated the **Google GenAI SDK** with Node.js (**fs**, **path**) to automate code analysis, traverse project files, and generate AI-powered fixes, secure code patches, and refactoring suggestions.

## CERTIFICATIONS

**Foundations of Cybersecurity:** Coursera (Google)

**Introduction to Cloud Computing:** SkillsBuild (IBM)

**DSA JAVA:** ApnaCollege

**Web Development:** Agratas Edutech (AICTE)

